INCH-POUND

MIL-DTL-32123B(OS)
28 January 2025
SUPERSEDING
MIL-DTL-32123A(OS)
18 October 2006

DETAIL SPECIFICATION

LINER, POLYBUTADIENE-BASED

This specification is approved for use by the Naval Surface Warfare Center Indian Head Division (NSWC IHD) and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers three types of polybutadiene-based liner materials.

1.2.1 Types.

a. Type I – CCU-22B/A and CCU-149/A impulse cartridges

b. Type II – CKU-5C/A rocket catapult

c. Type III – CKU-7A/A, CKU-10A/A, and CKU-11A/A rocket catapults; CKU-10A/A and CKU-11A/A catapult grains; and MK122 Mod 1 parachute deployment rocket motor (PDRM)

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-C-85498 - Curing Agents, Dimeryl-Di-Isocyanate and Isophorone Di-Isocyanate
NOTE: MIL-C-85498 is inactive for new design and will be eventually replaced.

(Copies of this document are available online at https://quicksearch.dla.mil.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

Comments, suggestions, or questions on this document should be addressed to Commander, Naval Sea Systems Command, ATTN: SEA 05S, 1333 Isaac Hull Avenue, SE, Stop 5160, Washington Navy Yard, DC 20376-5160 or emailed to CommandStandards@navy.mil, with the subject line "Document Comment". Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

AMSC N/A FSC 1377

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NAVAL SURFACE WARFARE CENTER (NSWC IHD) DOCUMENTS

WS 19738 - Antioxidant, T-Butylphenol type

WS 20700 - Hydroxyl Terminated Polybutadiene

WS 20883 - Triphenylbismuth

NAVAL AIR SYSTEMS COMMAND (NAVAIR) DRAWINGS

673AS131 - Liner Filler

(Copies of these documents are available from Commander, Indian Head Division, Naval Surface Warfare Center, CAD/PAD Data Management Branch (Code E26), 4393 Benson Road, BLDG 1557, Indian Head, MD 20640-5092. Refer questions, inquiries, or problems to nswc_iheodtd_dmo@navy.mil.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

CHEMICAL PROPULSION INFORMATION AGENCY (CPIA)

Publication No. 21 - JANNAF Solid Propellant Physical Behavior Manual

(Application for copies should be addressed to Chemical Propulsion Information Agency, The Johns Hopkins University Applied Physics Laboratory, Johns Hopkins Road, Laurel, MD 20707.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Material. The material shall be a polybutadiene-based composition as described herein.

3.2 Chemical composition. The chemical composition and ingredients shall be in accordance with table I. The ingredients shall be in accordance with the referenced documents.

3.3 Modification of liner formulation. The properties of the liner may be varied by changing the ratio of curing agent, dimeryl-di-isocyanate (DDI) or isophorone-di-isocyanate (IPDI), to hydroxyl-terminated polybutadiene (HTPB). The amounts of curing agent and HTPB shall be determined by adjusting the ratio of chemical equivalents of the curing agent to the chemical equivalents of the HTPB. The value of this ratio shall nominally be 1.5/1.0 based on the functional equivalence values from the vendor's ingredient certifications. The total weight percent of the HTPB, DDI, and antioxidant for the type I material shall remain constant at 70 percent. The total weight percent of the HTPB, curing agent, and antioxidant for the type II and III materials shall remain constant at 80 percent.

TABLE I. Liner ingredients and composition.

Ingredients	Specification/type	Percent by weight type I	Percent by weight type II	Percent by weight type III
HTPB	WS 20700, type I	51.60±0.5160	58.96±0.5896	70.35±0.7035
Antioxidant	WS 19738	0.50±0.0050	0.59±0.0059	0.59±0.0059
Liner filler	673AS131	30.00±0.3000	20.00±0.2000	20.00±0.2000
Dimeryl-di-isocyanate	MIL-C-85498, type I	17.90±0.1790	20.45±0.2045	-
Isophorone-di-isocyanate	MIL-C-85498, type II	-	-	9.03±0.0903
Triphenylbismuth	WS 20883	-	-	0.03±0.0003

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3.4 Liner bond strength.

3.4.1 Propellant to liner. Propellant to liner bond strength testing (see 4.3.3) shall be required before a new liner ingredient lot set or liner formulation modification (see 4.2.1) is used. The propellant to liner bond shall exhibit cohesive failure in the propellant or cohesive failure in the propellant at the liner interface when tested at 77±5 °F.

3.5 Liner cure. The type I and II liners shall be cured at 140±5 °F for a minimum of 48 hours. The type III liner shall be cured at 140±5 °F for a minimum of 16 hours (see 6.4).

3.6 Workmanship. The material shall be uniform in consistency and free from contamination, foreign matter, or any other defect that would prevent its use for the purpose intended.

4. VERIFICATION

4.1 Inspection conditions. Unless otherwise specified (see 6.2) all inspections shall be performed as specified in the applicable test method of 4.3.

4.2 Conformance inspections. Conformance inspection shall consist of the tests specified in 4.3.1, 4.3.2, and 4.3.3. When specified (see 6.2), a report shall be prepared giving the results obtained for all inspections and tests performed and a certified statement that meets all requirements of this specification.

4.2.1 Inspection lot. An inspection lot shall consist of a single liner mix.

4.2.1.1 Liner mix. A liner mix may consist of a liner premix consisting of the HTPB, antioxidant, and carbon black with the curative, DDI or IPDI, and triphenylbismuth added after the premix is complete.

4.2.1.2 Ingredient lot set. The ingredient lot set shall be a set of the materials as specified in table 1 for each type of liner material, where each of the materials is from a single material lot that is acceptable under the quality provisions of this specification. If one or more of the material lots is changed from an established ingredient lot set, this constitutes a new ingredient lot set. As long as the formulation (the weight percent of the individual ingredients) established with a new ingredient lot set is maintained, no additional testing (see 3.4.1) is required. The testing of 4.3.3 shall be performed if any formulation modification is required with an ingredient lot set or if the ingredient lot set changes.

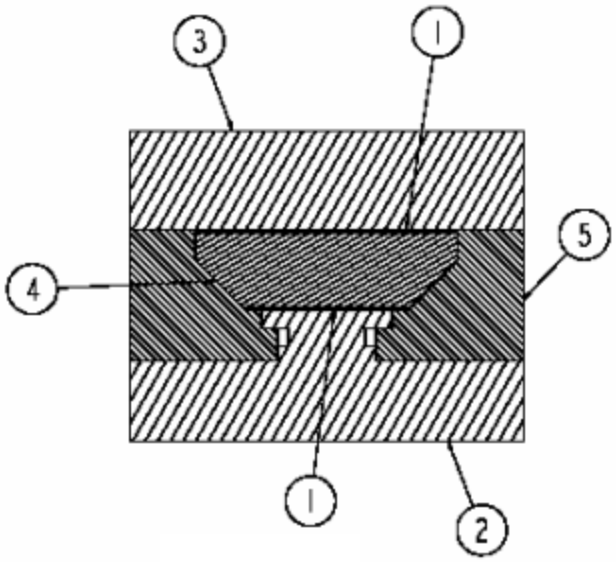
4.3 Test methods. Unless an alternative inspection method is approved by the Technical Authority (see 6.5) prior to use in testing, tests shall be performed using the apparatus and procedures specified herein.

4.3.1 Ingredient weight. Each liner ingredient shall be weighed to determine compliance with 3.2 prior to mixing. Weights shall be measured with equipment having an accuracy of ±1 percent of the ingredient weight calculated from the percentages specified in the applicable table (see 3.2).

4.3.2 Raw material. Each liner material shall be tested to determine compliance with 3.2.

4.3.3 Bond strength. The liner bond strength specimens shall be prepared and tested in accordance with CPIA Publication No. 21. In lieu of the machined specimen, a cast specimen as shown in figure 1 shall be used. Note that the plug (number 2) and cap (number 3) can be changed to best adapt to the available tension tester. A pull rate of 0.2 inches per minute shall be required. A minimum of four samples shall be tested to determine compliance with 3.4.1.

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FIND NUMBER	DESCRIPTION
1	LINER
2	PLUG
3	CAP
4	PROPELLANT
5	MOLD

FIGURE 1. Cast test specimen for conical bond in tension (C-BIT).

4.3.3.1 Liner test specimen. The propellant from the propellant mix used to cast the test specimens shall meet the propellant mechanical properties specified in the applicable propellant specification before the liner bond test is completed. If the propellant does not meet the mechanical properties specified in the applicable propellant specification, all bond testing completed with that propellant mix shall be deemed invalid.

a. The C-BIT test specimen shall be made using a uniformly distributed amount of liner on the 1-inch-diameter surface of the plug (number 2) that emulates the thickness (±20 percent) of the liner used to line the motor tubes or sleeves.

b. The cap (number 3) in figure 1 shall have liner coating it to provide adhesion to propellant.

c. The cap (number 3) and plug (number 2) shall be cured at the same temperature and time as the lined propellant tubes (60±12 hours at 140±5 °F for the type I and II liners and 20±4 hours at 140±5 °F for the type III liner).

d. Propellant shall be cast into the assembly assuring that the cap (number 3) and plug (number 2) are axially aligned.

4.3.3.2 BC-22 propellant bond strength exception. The BC-22 propellant shall exhibit cohesive failure in the propellant or at the liner interface when tested at 165±5 °F, 77±5 °F, and -65±5 °F. The BC-22 propellant to liner bond test result shall meet a minimum cohesive bond failure, as seen in figures 2, 3, and 4, in addition to meeting the required bond strength. The C-BIT bond tests in figures 2, 3, and 4 are from the MK122 Parachute Deployment Rocket Motor Mod 1 (PDRM Mod 1) qualification using BC-22 propellant. As demonstrated in figures 2, 3, and 4, the minimum required cohesive failure in the propellant for the PDRM Mod 1 shall be a very faint tint of red propellant on top of the liner. Figures 2, 3, and 4 demonstrate the acceptable cases for three temperature ranges.

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FIGURE 2. C-BIT test result at 165 °F.



FIGURE 3. C-BIT test result at 77 °F.



FIGURE 4. C-BIT test result at -65 °F.

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5. PACKAGING

This section is not applicable to this specification.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. Liner materials in accordance with this specification are intended to be used as the liner between the case or tube wall and the propellant of various cartridge and rocket motor devices. The liner was designed for use in military cartridge and rocket motor devices, and thus has no commercial application.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Inspection conditions, if other than as specified (see 4.1).

c. Whether a conformance test report is required (see 4.2)

6.3 Manufacturing conditions. This material has been manufactured (processed, applied, and cured) successfully under the following conditions.

a. Humidity: 50 grains of water per pound of dry air, maximum.

b. Temperature: 70 to 90 °F.

6.4 Liner cure test. Liner cure can be determined as follows:

a. Using a nitrile rubber glove, touch the surface of the liner with a gloved hand.

b. Liner can be tacky but should not stick to or come off onto the glove.

6.5 Technical Authority. The Technical Authority has the authority, responsibility, and accountability to establish, monitor, and approve technical standards, tools, and processes in conformance to higher authority policy, requirements, architectures, and standards. NAVSEA 05S may be contacted to determine the appropriate Technical Authority for any given requirement related to equipment and systems of U.S. Navy ships and submarines.

6.6 Subject term (key word) listing.

Cartridges
HTPB
Rocket motors

6.7 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

CONCLUDING MATERIAL

Preparing activity:
Navy – OS
(Project 1377-2025-001)

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